Improving X-Ray Instance Segmentation with Point-Based Mask Refinement

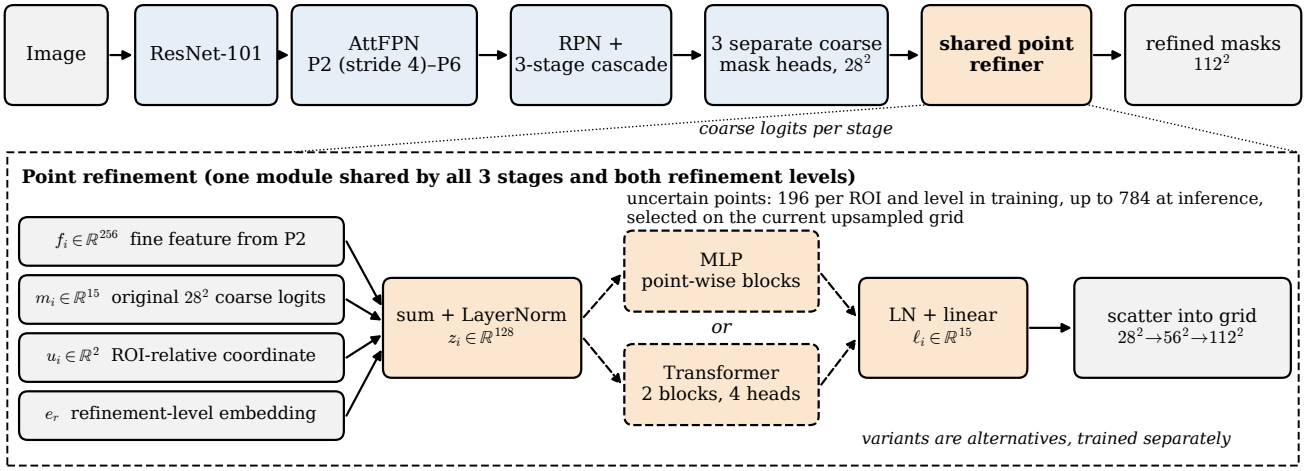


Figure 1: Our framework. Three separate cascade mask heads give coarse 28^2 masks; one shared point refiner updates uncertain locations at 56^2 and 112^2. MLP and transformer are alternative, separately trained variants, detailed in Fig. 2.

1 Method

Region-based detectors predict X-ray instance masks on a coarse 28×28 grid, which limits accuracy for thin and overlapping prohibited items. We keep a strong detector and improve its masks at selected points (Fig. 1). The parent is a three-stage Cascade Mask R-CNN with ResNet-101 and AttFPN; P2 denotes the stride-4 AttFPN map. Each stage has its own mask head producing 28×28 logits for 15 classes. One shared refiner upsamples these logits to 56^2, selects the most uncertain points on the current grid, refines them, and repeats at 112^2. Each point becomes a token

$$z_i = \mathrm{LN}(E_f(f_i) + E_m(m_i) + E_p(u_i) + e_r) \in \mathbb{R}^{128}, \quad (1)$$

where $f_i \in \mathbb{R}^{256}$ is the P2 feature, $m_i \in \mathbb{R}^{15}$ the original coarse logits, $u_i \in \mathbb{R}^2$ the ROI coordinate and e_r a level embedding. With Z the tokens of one region,

$$\ell_i = W_o\, G(Z)_i + b_o \in \mathbb{R}^{15}, \quad (2)$$

where G is two pre-LayerNorm residual blocks and a final LayerNorm, using a point-wise MLP (MLP variant) or four-head self-attention within the region (transformer variant); Fig. 2 shows both as block diagrams. The logits ℓ_i replace the selected grid values. Training adds a point loss to the detector loss,

$$L = L_{\text{base}} + \sum_{s=1}^{3} w_s\, \tfrac{1}{2}\big(L^{s,56}_{\text{point}} + L^{s,112}_{\text{point}}\big), \quad w = (1, 0.5, 0.25), \quad (3)$$

a binary cross-entropy on the ground-truth class channel against targets sampled from the ground-truth masks. The transformer was our proposed variant; the MLP was added as a matched control. They add 458,383 and 458,127 parameters respectively.

2 Evaluation Protocol

Official PIXray split (3,560 training and 1,486 test images, 4,508 test instances, 15 classes), annotations unchanged. All methods train 60 epochs with AdamW (learning rate 10^{-4}, effective batch 16) and one seed, and the final-epoch checkpoint is evaluated; the refinement variants differ from the parent only in the ROI head. We report COCO box and mask AP over IoU 0.50:0.05:0.95, AP_{50}, AP_{75}, size-specific AP and AR@100 (score threshold 0.05), per-class mask AP, and mask IoU, Dice and boundary IoU on objects matched by both methods of a pair.

3 Results

The MLP variant achieves the highest observed mask AP, 71.50%, adding about 0.46 million parameters (Table 1). It improves mask AP by +2.38 pp over the parent, +2.28 pp over the bilayer baseline, +10.08 pp over Occ and +2.51 pp over the PIDray adaptation, and leads mask AP_{75} and mask AR@100. The transformer variant gains +0.95 pp; the MLP variant outperforms it by +1.43 pp and at all 10 IoU thresholds in this single-seed run.

Gains concentrate on mask quality. The MLP variant improves mask AP_S by +6.20 pp, AP_M by +1.57 pp and AP_L by +4.10 pp over the parent (Table 2), while box AP changes by −0.21 pp. The gain rises from +0.39 pp at IoU 0.50 to +5.00 pp at IoU 0.95 (Fig. 4): refinement produces tighter masks rather than more detections, while the bilayer and PIDray baselines stay close to the parent.

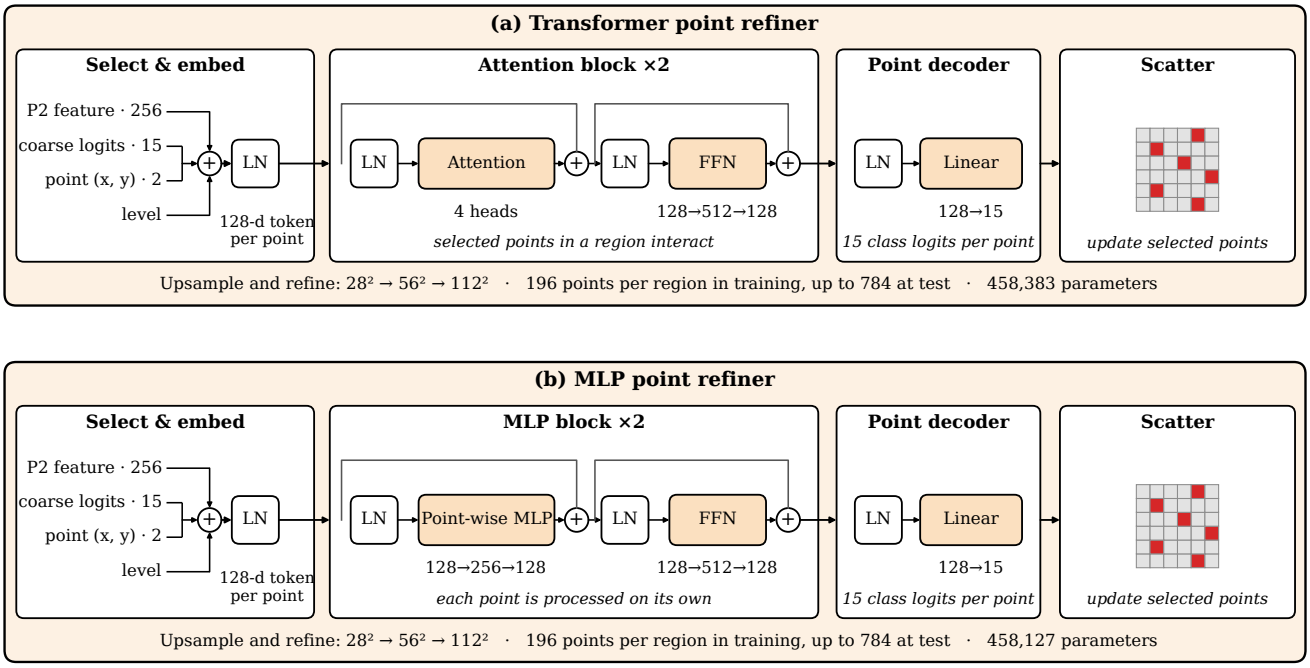


Figure 2: Inside the two point refiners. Both select and embed the same points, use the same feed-forward layer, decoder and scatter update, and differ only in one block: (a) attention lets the selected points of a region interact, while (b) the MLP processes each point on its own.

Table 1: PIXray official test split. Bold: best observed value. Δ: difference from the parent baseline in percentage points. One training seed. The bilayer baseline also uses a recall-enhanced RPN and auxiliary occluder masks; Occ uses SAM ViT-L and auxiliary occluder masks, with incompletely recorded prediction provenance. Baselines are adapted implementations.

Method	Params (M)	Box AP ↑	Mask AP ↑	Mask AP_{75} ↑	Mask AR@100 ↑	Δ Mask AP
Cascade Mask R-CNN + AttFPN	97.1	80.21	69.12	79.30	72.23	—
Cascade + Bilayer Mask Head	105.2	**80.23**	69.22	79.13	72.41	+0.10 pp ↑
Occ Adaptation	336.7	73.70	61.43	69.07	65.06	−7.70 pp ↓
PIDray R0/R1 Adaptation	98.7	80.12	68.99	79.14	72.14	−0.13 pp ↓
Point Refinement—Transformer (Ours)	97.6	80.01	70.07	78.62	73.16	+0.95 pp ↑
Point Refinement—MLP (Ours)	97.6	80.00	**71.50**	**81.41**	**74.42**	**+2.38 pp** ↑

Table 2: Detailed COCO metrics on the PIXray test split (%). All higher is better; bold marks the best observed value. AP_{50}/AP_{75} use one IoU threshold; AP_S/AP_M/AP_L group ground-truth instances by the annotation area field, which stores box area in PIXray. Box and mask AR@100 use at most 100 detections per image.

	Box ↑			Mask ↑			
Method	AP_{50}	AP_{75}	AR@100	AP_{50}	AP_S	AP_M	AP_L
Cascade Mask R-CNN + AttFPN	94.52	89.17	82.92	93.53	38.70	57.23	68.45
Cascade + Bilayer Mask Head	94.93	**89.71**	**82.99**	93.80	37.37	56.82	68.45
Occ Adaptation	**95.18**	85.34	77.48	91.80	19.97	45.44	63.66
PIDray R0/R1 Adaptation	94.92	89.48	82.88	93.39	41.25	56.58	69.22
Point Refinement—Transformer (Ours)	94.69	89.35	82.73	93.32	42.02	57.31	70.44
Point Refinement—MLP (Ours)	94.85	89.16	82.68	**93.92**	**44.90**	**58.80**	**72.55**

4 Per-Class and Instance-Level Analysis

Every class improves. The MLP variant raises mask AP over the parent in 15 of 15 classes and gives the best score in 13 of them; the transformer variant wins the remaining 2 (Table 3). The hardest classes (Dart, Saw_blade, Knife and Screwdriver) gain +2.96 pp on average, but Dart remains the weakest at 36.87%.

Masks are tighter on the same objects. On instances matched by both methods (Table 4), the MLP variant raises mask IoU by +1.07 pp, Dice by +0.65 pp and boundary IoU by +1.17 pp over the parent, improving

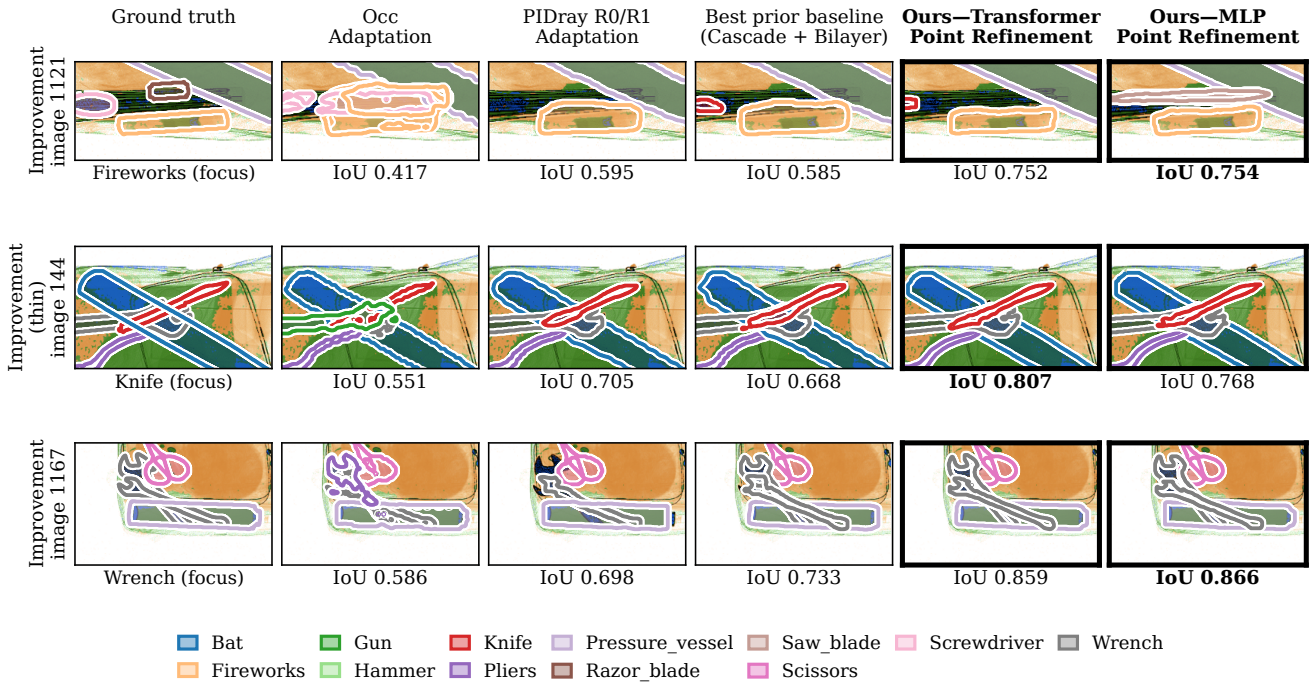


Figure 3: Illustrative improvement cases on identical crops, selected where both of our variants exceed all three baselines (row 2 from the hardest categories); not average performance. IoU is each method's best same-category mask IoU with the focus instance; every best mask is displayed (score ≥ 0.30).

Table 3: Per-class mask AP (%), classes sorted from hardest to easiest by mean AP over the six methods. Bold: best per class. Transf./MLP: our refinement variants.

Class	Parent	Bilayer	Occ	PIDray	Transf.	MLP
Dart	35.9	36.8	19.4	35.6	36.2	**36.9**
Saw_blade	44.9	45.3	37.1	45.0	42.9	**51.0**
Knife	55.2	54.6	44.2	54.9	54.8	**56.8**
Screwdriver	58.9	59.8	50.7	58.9	61.3	**62.1**
Pliers	59.8	60.0	49.7	60.1	59.8	**64.1**
Lighter	67.5	67.4	56.7	67.6	67.7	**68.3**
Razor_blade	68.7	68.9	60.0	68.8	68.4	**69.6**
Wrench	70.3	69.5	65.7	70.2	70.9	**73.0**
Scissors	74.4	74.6	61.3	74.2	75.9	**76.7**
Fireworks	75.9	76.0	72.1	75.1	**77.8**	76.7
Battery	79.3	79.1	73.4	78.9	**79.8**	79.7
Gun	84.0	83.8	79.3	83.2	84.9	**85.5**
Hammer	84.9	85.1	78.0	84.7	85.1	**86.0**
Pressure_vessel	88.1	87.9	85.0	88.2	90.8	**91.2**
Bat	88.8	89.6	88.9	89.4	94.6	**94.9**
Best in class	0	0	0	0	2	13

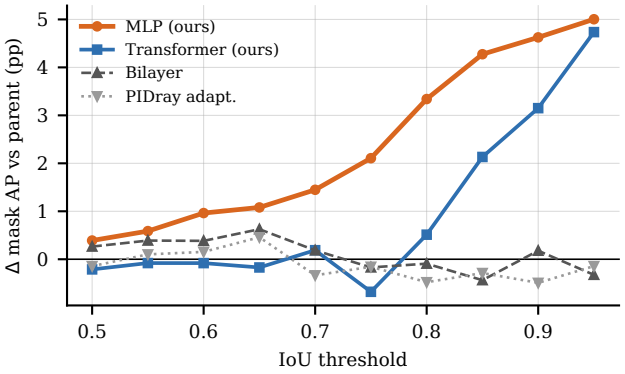


Figure 4: Mask AP gain over the parent at each IoU threshold.

70.8% of instances. Fig. 3 shows more complete fireworks, knife and wrench masks from both of our variants than from any baseline.

Where refinement does not help. Table 5 lists every metric. The MLP variant lowers 5 of 14 metrics, all of them box metrics, by at most 0.31 pp: refinement changes masks, not boxes. The transformer variant lowers 6 (Box AP, Box AP_S, Box AP_L, Box AR@100, Mask AP_{50} and Mask AP_{75}), falls below the parent at IoU 0.50, 0.55, 0.60, 0.65 and 0.75, and in 4 classes (Knife, Pliers, Razor_blade and Saw_blade). Even for the MLP variant, 29.2% of matched instances get a lower mask IoU than with the parent, and the transformer beats the MLP on 39.1% of the instances both match.

5 Conclusion

Point-based refinement improves X-ray instance segmentation. Our MLP variant reaches the highest observed mask AP, 71.50% (+2.38 pp over its parent), improving every class and object size, most at strict IoU thresholds.

Table 4: Instance-level mask quality, in percentage points, on ground-truth instances matched by *both* methods (score ≥ 0.5, one-to-one, same class, box IoU ≥ 0.5). Dice $= 2|P \cap G|/(|P| + |G|)$; boundary IoU uses a band of 2% of the image diagonal at original resolution. Cov.: % of all ground-truth instances in the pair; Impr.: % of pairs with higher IoU.

Pair	Cov.	ΔIoU	ΔDice	ΔB-IoU	Impr.
MLP vs Parent	92.3	+1.07	+0.65	+1.17	70.8
Transf. vs Parent	92.7	+0.39	+0.20	+0.47	61.2
MLP vs Transf.	92.3	+0.68	+0.45	+0.69	60.9
MLP vs Bilayer	92.0	+1.12	+0.68	+1.24	71.6

Table 5: Change of every COCO metric relative to the parent (percentage points; ↑ increase, ↓ decrease). Mask metrics improve for the MLP variant; box metrics and several transformer mask metrics do not.

Metric	Transformer	MLP
Box AP	−0.20 ↓	−0.21 ↓
Box AP_{50}	+0.17 ↑	+0.33 ↑
Box AP_{75}	+0.19 ↑	−0.01 ↓
Box AP_S	−2.27 ↓	+3.26 ↑
Box AP_M	+0.11 ↑	−0.31 ↓
Box AP_L	−0.27 ↓	−0.22 ↓
Box AR@100	−0.19 ↓	−0.24 ↓
Mask AP	+0.95 ↑	+2.38 ↑
Mask AP_{50}	−0.21 ↓	+0.39 ↑
Mask AP_{75}	−0.68 ↓	+2.11 ↑
Mask AP_S	+3.32 ↑	+6.20 ↑
Mask AP_M	+0.08 ↑	+1.57 ↑
Mask AP_L	+2.00 ↑	+4.10 ↑
Mask AR@100	+0.93 ↑	+2.19 ↑